For face,

AFGO : $\hat{n} = -k$, $z = 0$ then.

$$\iint_{AFGO} \vec{F} \cdot \hat{n}\, ds = \int_0^1 \int_0^1 (-y^2 \hat{j}) \cdot (-k)\, dx\, dy = 0$$

Adding $\iint_S \vec{F} \cdot \hat{n}\, ds = 2 + 0 + (-1) + 0 + \frac{1}{2} + 0$

$$= \frac{3}{2}$$

Green's Theorem

(M)

State and prove Green's theorem in the xy - plane.

Statement: If R is a closed region of the xy plane bounded by a simple closed curve C and if M and N are continous function of x and y having continous derivatives in R, then.

$$\oint_C M\, dx + N\, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial N} \right) dx\, dy$$

where C is traversed in the positive directions.

problm: Verify Green's theorem in the plane for $\oint_C (xy + y^2)\, dx + x^2\, dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

Soln: Given $y = x$ & $y = x^2$

$\therefore x = x^2$

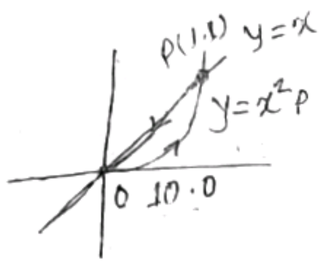
or, $(x^2 - x) = 0$

or, $x(x-1) = 0$

$\therefore x = 0$ or, $x = 1$

when $x = 0$ then $y = 0$

when $x = 1$ then $y = 1$



There fore the given curve interset at (0.0) and (1,1) The positive direction in traversing C is as show in the adjacend figure.

Along $y = x^2$ the line integral become,

$$\int_0^1 \{(x \cdot x^2 + x^4)\, dx + x^2 \cdot 2x\, dx\}$$

$$= \int_0^1 (x^3 + x^4 + 2x^3)\, dx$$

$$= \int_0^1 (3x^3 + x^4)\, dx$$

$$= \left[3\frac{x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$= \frac{3}{4} + \frac{1}{5}$

$= \frac{15+4}{20}$

$= \frac{19}{20}$

Along $y=x$ the line integral become,

$$\int_1^0 \{(x \cdot x + x^2)\,dx + x^2\,dx\}$$

$$= \int_1^0 (x^2 + x^2 + x^2)\,dx$$

$$= \int_1^0 3x^2\,dx = 3 \cdot \frac{x^3}{3}\Big|_1^0 = x^3$$

$$= (0-1)$$

$$= -1$$

Then the required line integral,

$$\frac{19}{20} - 1$$

$$= -\frac{1}{20} \quad \text{Ans:}$$

Where $M = xy + y^2$ & $N = x^2$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \iint_R \left\{\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(xy+y^2)\right\} dx\,dy$$

$$= \iint_R \{2x - (x+2y)\}\,dx\,dy$$

$$= \int_{x=0}^{1} \int_{y=x^2}^{y=x} (x-2y)\,dy\,dx$$

$$= \int_{x=0}^{1} \left[xy - 2\frac{y^2}{2}\right]_{y=x^2}^{y=x} dx$$

$$= \int_{x=0}^{1} \{x^2 - x^2 - (x^3 - x^4)\}\,dx$$

$$= \int_{x=0}^{1} (x^4 \cdot x^3)\,dx$$

$$= \left[\frac{x^5}{5} - \frac{x^4}{4}\right]_0^1$$

$$= \left[\frac{1}{5} - \frac{1}{4}\right]$$

$$= \frac{4-5}{20}$$

$$= \frac{-1}{20} \quad \text{Ans:-} \quad [\text{Hence the theorem is verifie}$$

problm: varify Green's theorem in the plane for $\oint (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$ where C is the boundary of the region defined by $y = \sqrt{x}$, $y = x^2$.

** Evaluate $\oint (x^2 - 2xy)\,dx + (x^2y + 3)\,dy$ around the boundary of the region defined by $y^2 = 8x$ and $x = 2$.

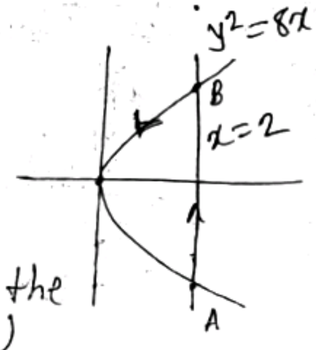
(a) directly (b) by using Green's theorem.

Soln: (a) Given that,

$$y^2 = 8x \text{ & } x = 2$$

$\therefore y^2 = 16$

$y = \pm 4$



$\therefore$ The ~~integr~~ intersecting point of the given curve is $(2, -4)$ and $(2, 4)$

along $x = 2$ the line integral equals,

$$\int_{-4}^{4} (4 - 4y)\cdot 0 + (4y + 3)\,dy$$

$$= \int_{-4}^{4} (4y + 3)\,dy$$

$$= \left[4\frac{y^2}{2} + 3y\right]_{-4}^{4}$$

$$= 2\cdot 16 + 12 - \{2\cdot 16 + 3\cdot(-4)\}$$

$$= 12 + 12$$

$$= 24$$

Along $y^2 = 8x$ the line integral equals,

$$x = \frac{y^2}{8}$$

$$dx = \frac{1}{8} 2y\,dy = \frac{1}{4} y\,dy$$

$$\int_{4}^{-4} \left\{\frac{y^2}{64} - 2\cdot\frac{y^2}{8}\cdot y\right\}\frac{1}{4} y\,dy + \left\{\frac{y^4}{64}\cdot y + 3\right\} dy$$

$$= \frac{1}{64}\int_{4}^{-4} \frac{y^5}{4} - 4y^4 + y^5 + 192)\,dy$$

$$= \frac{1}{64}\int_{4}^{-4} \left(\frac{5}{4} y^5 - 4y^4 + 192\right) dy$$

$$= \frac{1}{64}\left[\frac{5}{4}\cdot\frac{y^6}{6} - \frac{4}{5} y^5 + 192y\right]_{4}^{-4}$$

$$= \frac{1}{64}\left[\frac{5}{4}\cdot\frac{4^6}{6} + \frac{4}{5} 4^5 - 192\cdot 4 - \frac{5}{4}\frac{4^6}{6} + \frac{4}{5} 4^5 - 192\cdot 4\right]$$

$$= \frac{1}{64}\left[\frac{8}{5} 4^5 - 192\times 8\right]$$

$$= \frac{128}{5} - 24$$

Then the required line integral.

$$24 + \frac{128}{5} - 24$$

$$= \frac{128}{5} \quad \text{Ans:-}$$

Date: 22-5-18

** Find the angle between the surface $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at the point $(2,-1,2)$

Soln: Let, $\phi_1 \equiv x^2+y^2+z^2-9=0$ ——(i)

and $\phi_2 \equiv x^2+y^2-z-3=0$ ——(ii)

A normal to (i) at $(2,-1,2)$ is,

$$\nabla\phi_1 = \frac{\partial\phi_1}{\partial x}\hat{i} + \frac{\partial\phi_1}{\partial y}\hat{j} + \frac{\partial\phi_1}{\partial z}\hat{k}$$

$$= 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$= 4\hat{i} - 2\hat{j} + 4\hat{k} \quad \text{at } (2,-1,2)$$

A normal to the surface (ii) at $(2,-1,2)$ is,

$$\nabla\phi_2 = \frac{\partial\phi_2}{\partial x}\hat{i} + \frac{\partial\phi_2}{\partial y}\hat{j} + \frac{\partial\phi_2}{\partial z}\hat{k}$$

$$= 2x\hat{i} + 2y\hat{j} - \hat{k}$$

$$= 4\hat{i} - 2\hat{j} - \hat{k} \quad \text{at } (2,-1,2)$$

The angle between the surface at the point is the angle between the normals to the surface at the point. Let θ be the angle between (i) & (ii)

$$\therefore \nabla\phi_1 \cdot \nabla\phi_2 = |\nabla\phi_1||\nabla\phi_2| \cos\theta$$

$$\therefore \cos\theta = \frac{(4\hat{i}-2\hat{j}+4\hat{k})\cdot(4\hat{i}-2\hat{j}-\hat{k})}{\sqrt{4^2+(-2)^2+4^2}\cdot\sqrt{4^2+(-2)^2+(-1)^2}}$$

$$= \frac{16+4-4}{\sqrt{16+4+16}\cdot\sqrt{16+4+1}}$$

$$= \frac{16+4-1}{\sqrt{36}\cdot\sqrt{21}}$$

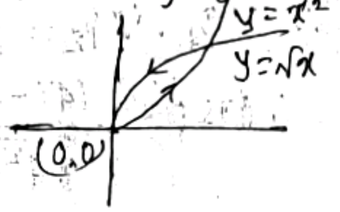
$$= \frac{16}{6\sqrt{21}}$$

$$= \frac{8}{3\sqrt{21}}$$

$$\therefore \theta = \cos^{-1}\frac{8}{3\sqrt{21}}$$

**100%

** verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region defined by $y = \sqrt{x}$, $y = x^2$.



Soln: Given that,

$y = \sqrt{x}$ & $y = x^2$

or, $y^2 = x$

or, $(x^2)^2 = x$

or, $x^4 - x = 0$

or, $x(x^3 - 1) = 0$

or, $x(x-1)(x^2 + x + 1) = 0$

$\therefore x = 0, 1$

Therefore the given two curves intersect at (0,0) and (1,1)

Along $y = x^2$, $dy = 2x\,dx$ from (0,0) to (1,1) then line integral become,

$$\int_0^1 (3x^2 - 8x^4)dx + (4\cdot x^2 - 6x\cdot x^2)2x\,dx$$

$$= \int_0^1 (3x^2 - 8x^4 + 8x^3 - 12x^4)dx$$

$$= \int_0^1 (3x^2 + 8x^3 - 20x^4)dx$$

$$= \left[\frac{3x^3}{3} + \frac{8x^4}{4} - \frac{20x^5}{5}\right]_0^1$$

$$= 1 + \frac{8}{4} - \frac{20}{5}$$

$$= 1 + 2 - 4$$

$$= -1 \quad \text{——— (i)}$$

Along $y = \sqrt{x}$. $dy = \frac{1}{2\sqrt{x}}dx$ from (1,1) to (0,0) H line integral become.

$$\int_1^0 (3x^2 - 8x)dx + (4\sqrt{x} - 6x\cdot\sqrt{x})\cdot\frac{1}{2\sqrt{x}}dx$$

$$= \int_1^0 (3x^2 - 8x + 2 - 3x)dx$$

$$= \int_1^0 (3x^2 - 11x + 2)dx$$

$$= \left[\frac{3x^3}{3} - \frac{11x^2}{2} + 2x\right]_1^0$$

$= 0 - (1 - \frac{11}{2} + 2)$

$= \frac{-2+11-4}{2}$

$= \frac{5}{2}$ ——— (ii)

Adding eqn (i) & (ii)

$-1 + \frac{5}{2} = \frac{-2+5}{2} = \frac{3}{2}$

We know Green's theorem,

$$\oint M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

According to our problem $M = 3x^2 - 8y^2$ and $N = 4y - 6xy$

$\therefore \frac{\partial M}{\partial y} = 0 - 16y$ and $\frac{\partial N}{\partial y} = 0 - 6y$

$$\therefore \oint (3x^2 - 8y^2)\, dx + (4y - 6xy)\, dx = \int_{x=0}^{1} \int_{y=x^2}^{y=\sqrt{x}} (-6y + 16y)\, dy\, dx$$

$$= \int_{x=0}^{1} \left[\frac{10y^2}{2}\right]_{y=x^2}^{y=\sqrt{x}} dx$$

$$= \int_0^1 (5x - 5x^4)\, dx$$

$$= \left[\frac{5x^2}{2} - 5\frac{x^5}{5}\right]_0^1$$

$$= \frac{5}{2} - 1$$

$$= \frac{3}{2}$$

** Find the work done in moving a particle once around a circle C in the xy plane, if the circle has centre at the origin and radius 3 and if the force field is given by

$\vec{F} = (2x - y + z)\vec{i} + (x + y - z^2)\vec{j} + (3x - 2y + 4z)\vec{k}$.

Soln: $W = \oint \vec{F} \cdot d\vec{r}$

$= \oint (2x - y)\, dx + (x + y)\, dy$

$= \int_0^{2\pi} (2(3\cos\theta) - 3\sin\theta)(-3\sin\theta\, d\theta) + (3\cos\theta + 3\sin\theta)\, 3\cos\theta\, d\theta$

$= \int_0^{2\pi} -18\cos\theta\sin\theta + 9\sin^2\theta + 9\cos^2\theta + 9\sin\theta\cos\theta\, d\theta$

$$= \int_0^{2\pi} (9 - 9\sin\theta\cos\theta)\, d\theta$$

$$= 9\int_0^{2\pi} d\theta - 9\int_0^{2\pi} \sin\theta\cos\theta\, d\theta$$

$$= 9\cdot\theta\Big|_0^{2\pi} - 9\int_0^{2\pi} \sin\theta\, d(\sin\theta)$$

$$= 9\cdot 2\pi - \frac{9\sin^2\theta}{2}\Big|_0^{2\pi}$$

$$= 18\pi - 2(\sin^2 2\pi - \sin^2 0)$$

$$= 18\pi \quad \underline{\text{Ans:-}}$$